



**Glass cloth base epoxy resin  
flame retardant copper clad laminate**

## NP-140F

### FEATURES

- High luminance of epoxy contrast with copper for laser type A.O.I.
- UV solder mask may be applied simultaneously to increase yields.
- High performance epoxy blended to achieve higher heat resistance

### PERFORMANCE LIST

| Characteristics                          |        | Unit              | Conditioning               | Typical Values                     | SPEC            | Test Method |
|------------------------------------------|--------|-------------------|----------------------------|------------------------------------|-----------------|-------------|
| Volume resistivity                       |        | MΩ-cm             | C-96/35/90                 | $5 \times 10^8 \sim 5 \times 10^9$ | $10^6 \uparrow$ | 2.5.17      |
| Surface resistivity                      |        | MΩ                | C-96/35/90                 | $5 \times 10^6 \sim 5 \times 10^7$ | $10^4 \uparrow$ | 2.5.17      |
| Permittivity 1MHZ                        |        | -                 | C-24/23/50                 | 4.5-4.7                            | 5.4 ↓           | 2.5.5.9     |
| Permittivity 1GHZ                        |        | -                 | C-24/23/50                 | 4.0-4.2                            | -               | 2.5.5.9     |
| Loss Tangent 1MHZ                        |        | -                 | C-24/23/50                 | 0.015-0.020                        | 0.035 ↓         | 2.5.5.9     |
| Loss Tangent 1GHZ                        |        | -                 | C-24/23/50                 | 0.012-0.014                        | -               | 2.5.5.9     |
| Arc resistance                           |        | SEC               | D-48/50+D-0.5/23           | 120 ↑                              | 60 ↑            | 2.5.1       |
| Dielectric breakdown                     |        | KV                | D-48/50                    | 60 ↑                               | 40 ↑            | 2.5.6       |
| Moisture absorption                      |        | %                 | D-24/23                    | 0.05-0.10                          | 0.35 ↓          | 2.6.2.1     |
| Flammability                             |        | -                 | C-48/23/50                 | 94V0                               | 94V0            | UL94        |
| Peel strength 1 oz                       |        | lb/in             | 288°C x10" solder floating | 9-13                               | 6 ↑             | 2.4.8       |
| Thermal stress                           |        | SEC               | 288°C solder dipping       | 200 ↑                              | 10 ↑            | 2.4.13.1    |
| Pressure cooker<br>(2 atm 120°C)         | 1/2 hr | SEC               | 288°C dipping              | 150↑                               | N/A             | -           |
|                                          | 1 hr   | SEC               | 288°C dipping              | 150↑                               | N/A             | -           |
|                                          | 2 hr   | SEC               | 288°C dipping              | 150                                | N/A             | -           |
| Flexural strength                        | LW     | N/mm <sup>2</sup> | A                          | 480-550                            | 415 ↑           | 2.4.4       |
|                                          | CW     | N/mm <sup>2</sup> | A                          | 415-480                            | 345 ↑           | 2.4.4       |
| Dimensional stability X-Y axis           |        | %                 | E-0.5/170                  | 0.005-0.030                        | 0.050 ↓         | 2.4.39      |
| Coefficient of thermal expansion         |        |                   |                            |                                    |                 |             |
| Z-axis before Tg                         |        | ppm/°C            | TMA                        | 50-60                              | N/A             | 2.4.24      |
| Z-axis after Tg                          |        | ppm/°C            | TMA                        | 220-320                            |                 |             |
| Glass transition temp                    |        | °C                | DSC                        | 135 ± 5                            | N/A             | 2.4.25      |
| Decomposition<br>Temperature (Td 5% W/L) |        | °C                | TGA                        | 310                                | N/A             | 2.4.24.6    |

**NOTE:**

Data shown are nominal values for reference only.

The average value in the table refers to samples of .062" 1/1.

**Noted: NP-140F product is NOT suitable for automobile relative application.**

Test method per IPC-TM-650



**補充：NP-140F 與 NP-140TL 基本物性對比：**

| 規格       |                |     | 1/1 1.6mm |          |
|----------|----------------|-----|-----------|----------|
|          |                |     | NP-140F   | NP-140TL |
| peeling  | NO. 1          |     | 12.60     | 13.48    |
|          | NO. 2          |     | 12.97     | 13.07    |
| PCT 2hr  | 吸水率 (%)        |     | 0.35      | 0.38     |
|          | 288°C 耐熱性(min) |     | 4 ↑       | 4 ↑      |
| Tg-DSC   | 不含銅            | Tg1 | 138.36    | 139.34   |
|          |                | Tg2 | 138.03    | 139.18   |
|          |                | △Tg | -0.33     | -0.16    |
| TMA(不含銅) | 260°C (min)    | 含銅  | 11        | 10       |
|          |                | 不含銅 | 16        | 18       |
|          | 288°C (min)    | 不含銅 | 2         | 2        |
| CTE(不含銅) | $\alpha 1$     |     | 40.75     | 47.53    |
|          | $\alpha 2$     |     | 238.9     | 270.0    |
|          | %              |     | 3.50      | 4.09     |
| CTE(不含銅) | Tg             |     | 128.50    | 129.03   |
| 彎曲強度     | 經向             |     | 633.56    | 625.88   |
|          | 緯向             |     | 509.75    | 483.90   |